

Dynamical systems and ergodic theory

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¹Based on the lessons by professor Marco Lenci

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Main definitions

Fundamental definition

Definition 0.1 A dynamical system (DS) over the set M (usually called phase space or reference space) is a family $\phi^t_{t \in \mathbb{G}}$ of transformations $\phi^t : M \rightarrow M$ with the (semi)group property:

$$\begin{aligned}\phi^0 &= id_M \\ \phi^{t+s} &= \phi^s \circ \phi^t\end{aligned}$$

Note: $\mathbb{G}$ must be a commutative semigroup with neutral element with respect to the "+" operation. In practice, for the contents of this course, $\mathbb{G} = \mathbb{R}, \mathbb{R}_0^+, \mathbb{Z}, \mathbb{N}$

Observation 0.1 Differential equations not depending on time derivatives satisfy the definition.

Usually the formalism is that, say x is the state now (meaning at $t = 0$), then the state at time t is given by $\phi^t(x)$.

Definition 0.2 If $\mathbb{G} = \mathbb{R}, \mathbb{R}_0^+$, then ϕ^t is called the flow of the DS. if, instead, $\mathbb{G} = \mathbb{Z}, \mathbb{N}$ usually you rename it as $T = \phi^1$, $n = t$ where T is called a map.

Observation 0.2 This is due to the fact that by just knowing T you can have enough information to know the future of the system, since

$$\phi^t = (\phi^1)^n = \underbrace{\phi^1 \circ \phi^1 \dots \circ \phi^1}_{n \text{ times}} = T^n$$

Note that from now on if n is an integer exponent it means to use the transformation n times. Also, if T is invertible (bijective) we denote, for $n \in \mathbb{Z}^+$:

$$T^{-n} = \underbrace{T^{-1} \circ T^{-1} \dots \circ T^{-1}}_{n \text{ times}}$$

Observe that $\forall n, m \in \mathbb{Z}$:

$$T^{n+m} = T^n \circ T^m$$

Notice that by having T be invertible, it means that by knowing it you can also go back in time, and as such you know everything about the system; we will see both bijective and not in this course. (here he drew a figure).

Chapter 1

Examples of simple dynamical systems

1.1 Important example: autonomous ODEs

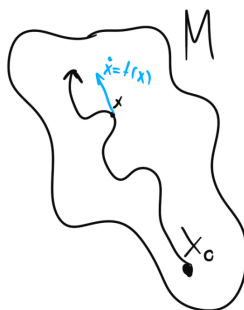


Figure 1.1: Representation of the quantities in an autonomous ODE.

The most obvious example of a dynamical system is given by an ODE, which takes the form of:

$$\dot{x} = f(x)$$

where $M \subseteq \mathbb{R}^d$, $f : M \rightarrow \mathbb{R}^d$.

We are looking at a curve of which we know the first time derivative, and we know said derivative to be a function of position, but not of time, as shown in Figure 1.1.

The aforementioned problem has, potentially, infinite solutions. What interests us are what are called Cauchy problems or, more commonly,¹ IVP, Initial

¹This is true for the whole world, apart from Italy and France, which only call those Cauchy problems. The latter has obvious reasons to do so. Italy, on the other hand, hasn't got so many.

Value Problems.

Definition 1.1 (Cauchy problem) A Cauchy problem is given by the following linear system:

$$\begin{cases} \dot{x} = f(x) \\ x(0) = x_0 \end{cases} \quad (1.1)$$

Cauchy's theorem tells us that under some conditions, which we assume, the solution exists, and it's unique. (those conditions being that $f(x)$ is a Lipschitz function).

We shall start with the classic ODE example: the harmonic oscillator.

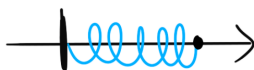


Figure 1.2: Representation of a harmonic oscillator.

Example 1.1 (Harmonic oscillator) The one dimensional harmonic oscillator is a physical system in which a mass is attracted to the origin by a force that depends on the distance to the origin itself, as shown in Figure 1.2. This is described by:

$$\ddot{x} = -w^2x \quad (1.2)$$

At first glance, it may look like it does not fit the definition of a Cauchy problem, since it is a second derivative instead of a first one. Turns out, all high order ODEs may be reduced to a single order one.

The trick is to rename the first derivative. in this case:

$$\begin{cases} \dot{x} = v \\ \dot{v} = -w^2x \end{cases}$$

Or, in matrix form:

$$\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} v \\ -w^2x \end{bmatrix} = f\left(\begin{bmatrix} x \\ v \end{bmatrix}\right)$$

We did increase the dimension of the phase space, but that does not get in the way of calling this a Cauchy problem.

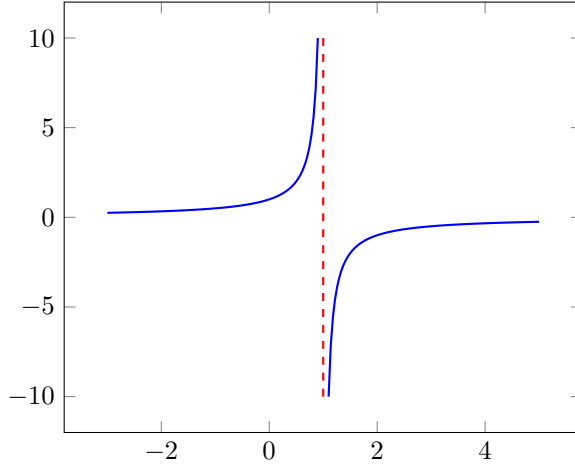
Other than uniqueness and existence, we will also be implicitly assuming all solutions to be global, i.e. defined $\forall t \in \mathbb{R}$, which is a much more restrictive condition. While finding a problem for which uniqueness and existence are not true is quite hard, the same is not true for a Cauchy problem without a global solution. Case in point:

Example 1.2 (of a non global solution)

$$\begin{cases} \dot{x} = x^2 \\ x(0) = 1 \end{cases}$$

where $M = \mathbb{R}$. Solving it by abusing math a bit:

$$\frac{dx}{dt} = x^2 \Leftrightarrow \frac{dx}{x^2} = dt \Leftrightarrow \int_1^x \frac{d\xi}{\xi^2} = \int_0^t d\tau \Leftrightarrow \left[-\frac{1}{\xi} \right]_1^x = t \Leftrightarrow 1 - \frac{1}{x} = t \Leftrightarrow x = \frac{1}{1-t}$$



As you can see, it blows up asymptotically at 1, so it is not a global solution.

We shall now see the concept of flow.

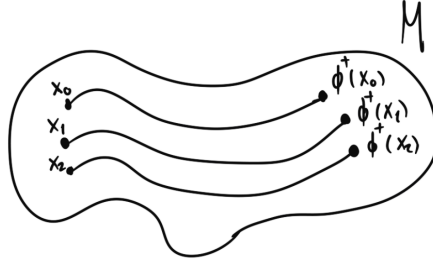


Figure 1.3: Representation of flow.

Definition 1.2 If $x(t; x_0)$ is the solution of Equation 1.1, the flow of such system is:

$$\phi^t(x_0) = x(t; x_0) \quad (1.3)$$

Proposition 1.1 $\phi_{t \in \mathbb{R}}^t$, as defined in Equation 1.3 satisfies the group property.

Proof. The group properties are two:

$$\phi^0 = id_M \quad (1.4)$$

$$\phi^{t+s} = \phi^s \circ \phi^t \quad (1.5)$$

Equation 1.4 is trivial, because by definition $\phi^0(x_0) = x_0 \forall x_0 \in M$.
The second property is a bit harder. First of all, by definition:

$$\phi^{t+s} = \phi^t \circ \phi^s \Leftrightarrow \phi^t(\phi^s(x_0)) = \phi^{t+s}(x_0) \forall x_0 \quad (1.6)$$

We define $y_1(t) = \phi^t(\phi^s(x_0))$ and $y_2(t) = \phi^{t+s}(x_0)$. We'll prove that $y_1(t)$ and $y_2(t)$ are both solutions of the following IVP:

$$\begin{cases} \dot{y}(t) = f(y(t)) \\ y(0) = \phi^s(x_0) \forall s \in \mathbb{R}, x_0 \in M \end{cases} \quad (1.7)$$

Because of Cauchy's theorem of uniqueness, if both solve the IVP, they must be equal, thus the proposition will be proved because of Equation 1.6.

The initial condition is true by definition:

$$\begin{aligned} y_1(0) &= \phi^0(\phi^s(x_0)) = \phi^s(x_0) \\ &\Leftrightarrow y_1(0) = y_2(0) = \phi^s(x_0) \forall s \in \mathbb{R} \\ y_2(0) &= \phi^{0+s}(x_0) = \phi^s(x_0) \end{aligned}$$

As for the other condition, for y_1 it's true by definition, first by the definition of y_1 itself, then by the definition of $\phi^t(x_0)$:

$$\begin{aligned} \frac{dy_1}{dt}(t) &= \frac{d\phi^t}{dt}(\phi^s(x_0)) = \frac{dx(t; \phi^s(x_0))}{dt} = \\ f(x(t; \phi^s(x_0))) &= f(\phi^t(\phi^s(x_0))) = f(y_1(t)) \end{aligned}$$

where x is the solution of the IVP over which the flow is defined, while $f()$ for which $\dot{x} = f(x)$ is true.

In the same way, by employing the definitions of y_2 and flow, aided by the chain rule, we may find that the temporal derivative of y_2 is equal to $f(y_2)$, where $f()$ is the same as the one for which that is true for y_1 :

$$\begin{aligned} \frac{dy_2}{dt}(t) &= \frac{d\phi^{t+s}}{dt}(x_0) = \frac{dx(t+s; x_0)}{dt} \frac{d(t+s)}{dt} = \\ f(x(t+s; x_0)) &= f(\phi^{t+s}(x_0)) = f(y_2(t)) \end{aligned}$$

So, y_1 and y_2 both solve the aforementioned IVP for every t , thus we proved the proposition. □

Let's now look a bit into the concept of flow. The flow behaves in such a way (this is why it's called a flow!) in which it never intercepts. It looks, if drawn in a stylized way, as a series of lines that never intercept, because of the uniqueness of the solution of the Cauchy problem, as shown in Figure 1.3.

Proposition 1.2 *We define two flows $\phi^{t_1}(x_1)$ and $\phi^{t_2}(x_2)$. If $t_1 \geq t_2$ and $x_1 \neq x_2$, they shall never intercept.*

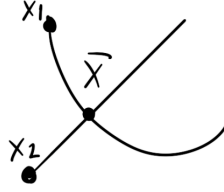


Figure 1.4: Representation of the hypothesis used to prove Proposition 1.2.

Proof. RAA² Assume that we have two trajectories that are completely distinct apart from one singular interception. Let's call the singular interception $\bar{x}$. The following is true by definition:

$$\bar{x} = \phi^{t_1}(x_1) = \phi^{t_2}(x_2)$$

But then, by definition of flow:

$$\phi^{-t_2}(\bar{x}) = \phi^{-t_2}(\phi^{t_2}(\bar{x})) = x_2$$

but at the same time:

$$\phi^{-t_2}(\bar{x}) = \phi^{-t_2}(\phi^{t_1}(x_1)) = \phi^{t_1-t_2}(x_1)$$

The hypothesis here that the interception is just one. Starting from said hypothesis, we found another interception, namely x_2 . This would mean, because of the uniqueness of the solution of the Cauchy problem, that the trajectories are exactly the same: but that is impossible, because we assumed that they were distinct apart from one singular point. $\square$

Example 1.3 (of global solution ascertained by visual inspection) Suppose M is a two dimensional domain and suppose you know that at the boundary the vector field f always points inside. And suppose you know that this vector field is smooth. Then you are assured that the solution is global, because you cannot escape.

Usually you can find a global solution because you can iteratively move inside small intervals.

²Reductio ad absurdum. I will use this to signal that I am proving by contradiction.

Chapter 2

Topological dynamics

Chapter 3

Ergodic theory

Chapter 4

Possible developments